

Judge Whether Data Supports a Claim

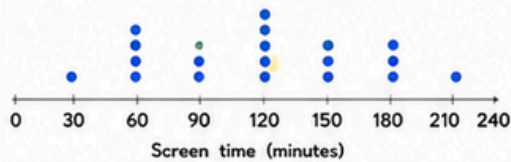
★ I can decide if data supports a claim and explain my reasoning using evidence.

SDA MS Step 10

Not checked yet

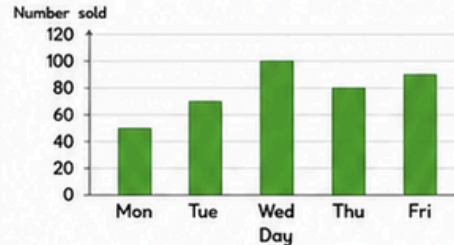
1 Let's Look A class collected data to test some claims.

A. Daily Screen Time (minutes)



Mean (average)	Median	Mode	Range
120 min	120 min	90, 120	210 min

B. Water Bottles Sold in a Week



Mean (average)	Median	Mode	Range
78	80	None	50

Talk About It

- What type of data display is used in A?
- What type of data display is used in B?
- What does the mean tell us?
- What does the range tell us?
- Why are different summary measures useful?

2 Let's Try Read each claim. Decide whether the data supports the claim. Explain your answer.

A. Claim

"The average daily screen time is less than 90 minutes."

Data: Daily Screen Time (minutes)

Mean	Median	Range
120 min	120 min	210 min

Does the data support the claim?

☐ Yes ☐ No

Explain your answer.

B. Claim

"Most students spend between 60 and 150 minutes on screens."

Data: Daily Screen Time (minutes)

Screen time (min)	Number of students
0-30	1
30-60	3
60-90	6
90-120	8
120-150	7
150-180	4
180-210	2
210-240	1
Total	32

Does the data support the claim?

☐ Yes ☐ No

Explain your answer.

C. Claim

"More water bottles were sold on Thursday than on Friday."

Data: Water Bottles Sold

Day	Number sold
Mon	50
Tue	70
Wed	100
Thu	80
Fri	90

Does the data support the claim?

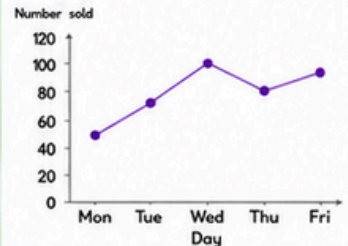
☐ Yes ☐ No

Explain your answer.

D. Claim

"Water bottles sold are increasing each day."

Data: Water Bottles Sold



Does the data support the claim?

☐ Yes ☐ No

Explain your answer.

3 Let's Do Use the data set to answer the questions.

E. Claim

"The school can reduce paper waste by at least 20% by using digital handouts."

Months	Paper used (kg)
Before (average per month)	100 kg
After (average per month)	76 kg

- What is the average paper used before? _____ kg
- What is the average paper used after? _____ kg
- By how many kilograms did paper use decrease? _____ kg
- What is the percentage decrease in paper use? _____ %
- Does the data support the claim? ☐ Yes ☐ No

Explain your answer.

Think About It

- What other information might you need to make a stronger decision about the claim?
- How could this data be collected more fairly?

4 Challenge Read the scenario and judge the claim.

A new sports drink claims: "Our drink improves endurance by at least 15% compared to water." A group of 20 students ran on a treadmill for 20 minutes. The table shows the distance (in metres) each group ran.

Distance Run in 20 Minutes (m)

Group	Mean	Median	Range	Minimum	Maximum
Sports Drink	2420	2410	420	2000	2420
Water	2100	2120	500	1700	2200

- By how many metres is the mean distance greater for the sports drink? _____ m
- What is the percentage improvement? _____ %
- Does the data support the claim? ☐ Yes ☐ No
- Explain your answer using evidence from the table.

My Learning Check

How did I go today?



I can do it!



Almost there!



I need more help.



This was hard.

Circle a face.

Parent Observation (optional)

Tick what you saw today.

- ☐ Interpreted data displays and tables
- ☐ Used summary measures (mean, median, range, mode)
- ☐ Judged whether data supports a claim
- ☐ Explained reasoning using evidence
- ☐ Needed support

Notes: _____

Key Words

claim – a statement that can be tested
evidence – information that supports a claim
mean (average) – total divided by number of values
median – the middle value when ordered
range – the difference between the largest and smallest values
percent – parts out of 100